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Job Insecurity: An Integrative Review and Agenda for Future Research

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Job insecurity reflects a threat to the continuity and stability of employment as it is currently experienced. Job insecurity has been the focus of increasing scholarly and popular attention in light of technological, economic, and political changes over the past few decades that have left many insecure about the future of their jobs. Yet, conceptual ambiguities exist; the literature remains fragmented; and there lacks an overarching framework through which to organize and reconcile findings. The goal of this article is to offer an integrative review and conceptual framework that addresses these challenges and provides the groundwork for future research. To that end, it proposes a definition of job insecurity that differentiates it from potential antecedents, moderators, and outcomes. The article addresses antecedents and introduces a typology of mechanisms and threat foci that links antecedents to job insecurity and suggests yet unexplored predictors. Furthermore, the framework developed here considers four overarching mechanisms—stress, social exchange, job preservation motivation, and proactive coping—through which job insecurity leads to various outcomes, and it highlights potential competing tensions inherent in individuals' responses. Finally, the framework introduces threat features, economic vulnerabilities, and psychological vulnerabilities as three overarching categories of variables that moderate reactions to job insecurity, and it identifies factors that contribute to each. In doing so, it suggests important levers through which to influence reactions to job insecurity; it helps explain variability in past research; and it provides a foundation for future work.

Keywords: *job insecurity; occupational health and safety; psychological contract; organizational change; labor market policies*

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The history of work is that of transformation. Over the past few decades, transformative technological, economic, and political changes surrounding work have left many insecure about the future of their jobs (Benach et al., 2014). In particular, technological advances, despite many benefits, have altered job tasks and demand for certain types of work (Kalleberg, 2011). Global competition has placed pressure on organizations to cut costs and adapt business practices (Ahean, 2012). Increasing ties to financial markets have exposed employers and employees to their ebb and flow (G. F. Davis, 2009). Against a backdrop of declining union power, it appears that stability and predictability in employment have been increasingly replaced by job insecurity (JI) as employers downsize, outsource labor, and demand that employees do more with less.

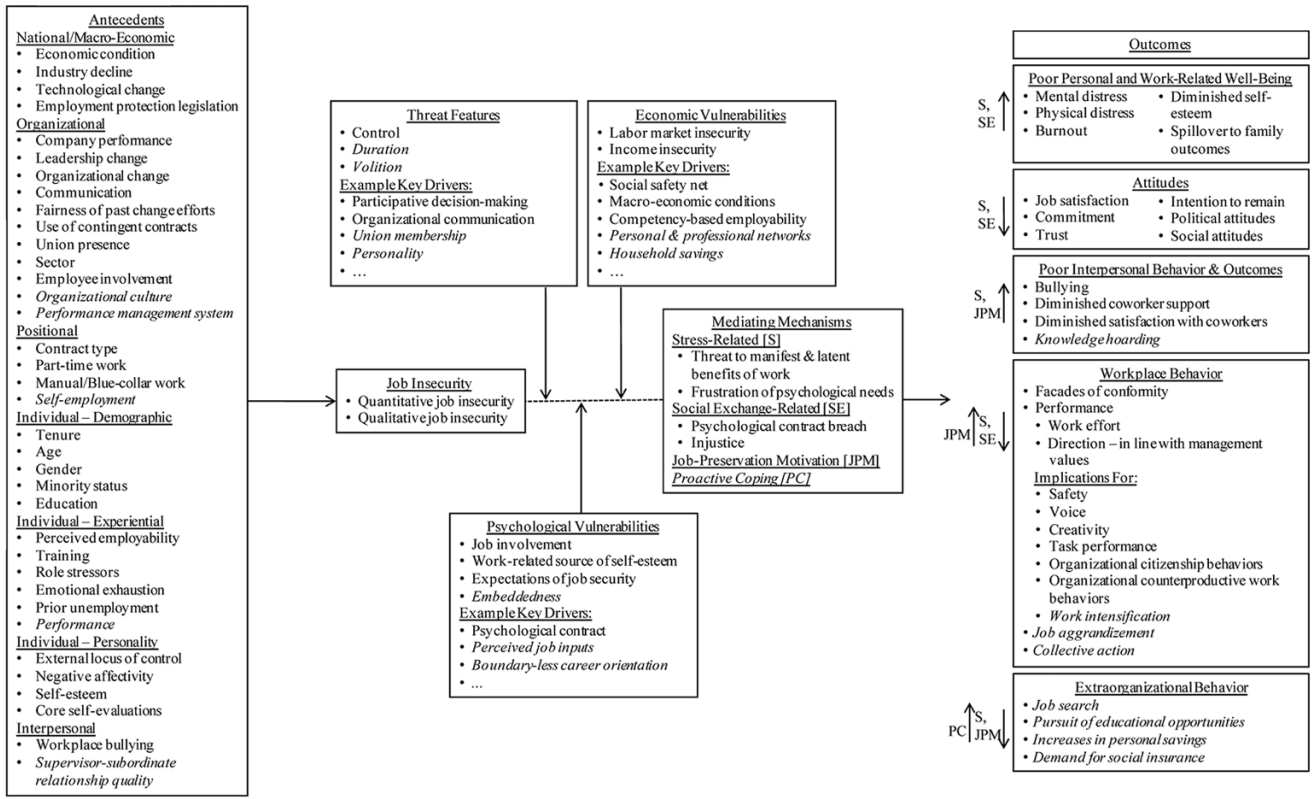
Not surprising, JI has become a topic of increasing scholarly and popular concern. JI has been characterized as a threat to population health (Burgard, Brand, & House, 2009), a potential mechanism behind health inequalities (Benach et al., 2014), and a tipping-point phenomenon both driving and resulting from organizational decline (Greenhalgh & Rosenblatt, 2010). Indeed, empirical evidence links JI to poor mental, physical, and work-related well-being; poor job attitudes; and decrements in performance, creativity, and adaptability (e.g., Cheng & Chan, 2008; De Witte, Pienaar, & De Cuyper, 2016; Ferrie et al., 2001; Niessen & Jimmieson, 2016; Probst, Stewart, Gruys, & Tierney, 2007). Yet, at the same time, some argue that job security creates complacency (Brockner, Grover, Reed, & Dewitt, 1992), and organizations and policy makers struggle with how to manage the trade-offs between organizational flexibility and employee job security (European Commission, 2007).

This article aims to move the JI literature forward in several important ways. First, it addresses issues of construct conceptualization and measurement and, in doing so, aims to provide future research with a clear conceptualization of JI and recommendations for measurement. Second, it consolidates and extends knowledge regarding antecedents of JI by offering a typology of mechanisms and threat foci that links antecedents to JI and suggests yet unexplored predictors. Third, the conceptual framework developed here addresses theoretical mechanisms underlying responses to JI and introduces several novel outcomes, including yet unexplored job preservation and proactive coping strategies. Finally, the framework introduces three overarching sets of moderating variables that exacerbate and attenuate employees' reactions to JI, and identifies factors that contribute to each. In doing so, it consolidates existing knowledge, explains variability in past work, and suggests levers through which reactions to JI can be shaped. The hope is that this framework provides a generative agenda for future research, as well as insights for management practice and for ongoing discussions regarding the consequences of an increasingly insecure world of work.

Conceptual Framework of JI

This review considers JI's antecedents, outcomes, and moderators within an overarching conceptual framework, displayed in Figure 1. The framework summarizes existing evidence regarding antecedent conditions and suggests new directions for identifying antecedents based on an underlying conceptual logic regarding how and why JI emerges. The framework coalesces and extends explanations for JI's effects by proposing four mechanisms through which JI affects outcomes—stress, social exchange, job preservation motivation, and proactive coping. Stress and social exchange mechanisms capture the meaning of threats; job

Figure 1
Conceptual Model of Antecedents and Outcomes of Job Insecurity



Note: Variables in italics and ellipses reflect unanswered questions for future work. The arrows on the left of the outcomes illustrate general patterns of corresponding and competing predictions made by the different mechanisms. Feedback loops are omitted for clarity.

preservation motivation and proactive coping respectively capture the ways by which individuals try to keep threats from coming to fruition or prepare themselves in case of eventual loss. Notably, these mechanisms offer corresponding and competing predictions and suggest yet unexplored outcomes of JI. Finally, in light of evidence that not all individuals are equally affected by JI, the framework proposes three overarching categories of variables that moderate reactions to JI by accentuating or attenuating these mechanisms. It also identifies key drivers, or sources, of these moderators. These moderating variables involve features around which threats may vary (i.e., elements of the stimulus), as well as ways in which individuals can be vulnerable to the negative effects of JI—economically (in terms of ease of replacing employment or income if lost) and psychologically (in terms of important identities, roles, and expectations tied to one's job or its security). Whereas economic vulnerabilities are tied to context (e.g., in terms of the labor market demand, social safety net), psychological vulnerabilities concern elements of the person, in particular the self.

The current framework differs from past treatments in several notable ways. First, the framework does not presume organizational events as the sole initiating causes of JI (Greenhalgh & Rosenblatt, 1984), nor does it restrict the study of JI to only those threats where the job itself (as opposed to the jobholder) is at risk (Jacobson & Hartley, 1991). Second, while existing treatments have been bounded within disparate theoretical perspectives—for example, Ashford, Lee, and Bobko (1989) operate within a psychological contract perspective, whereas Sverke, Hellgren, and Näswall (2002) take a stress perspective—the current framework transcends these perspectives to present a more comprehensive view of the ways in which individuals view and respond to JI and the potential competing tensions in their responses. Third, the framework does not specify outcomes as inherently short- or long-term, in contrast to Sverke et al. (2002), but calls for systematic attention to change over time. Finally, while the literature has largely taken a piecemeal approach toward moderators of JI-outcome relationships, the current framework provides a systematic conceptual basis to organize moderators and understand their effects. Note that while personality is referenced in a general sense in this review, a complete discussion is beyond the scope of this article. Directions for future research are displayed in italics in the figure and discussed throughout.

Defining JI

The term *job insecurity* is used in a variety of ways (see Table 1 for illustrative definitions). This poses a challenge, as the building of a coherent and practically useful body of knowledge surrounding JI requires a clear conceptualization of the construct. The current paper defines JI as *a perceived threat to the continuity and stability of employment as it is currently experienced.* With this regard, *quantitative JI* denotes perceived threats to the job as a whole (De Witte, 1999; Sverke & Hellgren, 2002), whereas the term *qualitative JI* denotes perceived threats to features of the job, in particular "threats of impaired quality in the employment relationship" (Hellgren, Sverke, & Isaksson, 1999: 182). This definition maintains and coalesces key elements from existing definitions, such as the notion of threat, the emphasis on perception, and the idea that JI can occur when one's job and job conditions are at risk. However, it omits elements such as expectations for security and involuntary nature of threats in order to distinguish JI from potential antecedents, moderators, and

Table 1
Sample Definitions of Job Insecurity

Reference	Definition
Greenhalgh & Rosenblatt (1984)	"Perceived powerlessness to maintain desired continuity in a threatened job situation"
C. V. Van Vuuren & Klandermans (1990)	"Concern about the future permanence of the job, or sometimes a concern about a significant deterioration in conditions of employment"
Jacobson & Hartley (1991)	"A discrepancy between the level of security a person experiences and the level she or he might prefer"
Heaney, Israel, & House (1994)	"An employee's perception of a potential threat to continuity in his or her current job"
De Witte (1999)	"An overall concern about the continued existence of the job in the future"
Hellgren, Sverke, & Isaksson (1999)	"Quantitative job insecurity refers to concerns about the future existence of the present job. Qualitative job insecurity pertains to perceived threats of impaired quality in the employment relationship, such as deterioration of working conditions, lack of career opportunities, and decreasing salary development."
Manski & Straub (2000; economics)	"Subjective probability of exogenous job destruction"
Sverke & Hellgren (2002)	"A fundamental and involuntary change concerning the continuity and security within the employing organisation."
Probst (2003)	"Job security [is] the perceived stability and continuance of one's job as one knows it"
De Cuyper, Bernhard-Oettel, et al. (2008)	"Employees' perceptions about potential involuntary job loss"
Klandermans, Hesselink, & Van Vuuren (2010)	"A concern about the future of one's job"
Burchell (2011; sociology)	"An employee's perception of the likelihood of the losing their job involuntarily in, say, the next six or twelve months"
Loi, Ngo, Zhang, & Lau (2011)	"A psychological state in which workers vary in their expectations of future job continuity within an organization"
De Cuyper, Mäkikangas, Kinnunen, Mauno, & De Witte (2012)	"Worker's perception or concern about potential involuntary job loss"
Schreurs, Van Emmerik, Günter, & Germeys (2012)	"Employees' feeling that their job is at risk or that they are likely to face job loss"
Keim, Landis, Pierce, & Earnest (2014)	"The overall apprehension of the continuing of one's job"
Vander Elst, De Witte, & De Cuyper (2014)	"The subjectively perceived and undesired possibility to lose the present job in the future, as well as the fear or worries related to this possibility of job loss"
Ellonen & Nätti (2015)	"Individual's evaluation of how likely it is that one will lose one's job in near future"
Vander Elst, De Cuyper, Baillien, Niesen, & De Witte (2016)	"The perceived threat of losing the current job in the near future"

outcomes and in line with criticisms levied by several authors (e.g., Probst, 2003; Vander Elst, De Cuyper, & De Witte, 2011). JI exists on a continuum from insecure to secure, where employees experience job security when they perceive that the continuity and stability of their jobs are unthreatened. This review focuses on JI as a global construct, instead of more narrowly focusing on its quantitative or qualitative components.

Defining Characteristics of JI

This definition points to several core elements of JI. While some of these elements have been variously articulated in other treatments (e.g., De Witte, 1999; T. Van Vuuren, Klandermans, Jacobson, & Hartley, 1991), it is worth highlighting them to be clear about what JI is and what JI is not. First, JI is a subjective experience (De Witte, 1999). This focus

on perceptions, though consistent with most conceptualizations, contrasts with the designation of jobs as objectively insecure—for example, based on contract type (e.g., temporary workers) or objective organizational circumstance (e.g., layoffs). Given the subjective nature of JI, two persons in the same objective situation may experience very different levels of JI (T. Van Vuuren et al., 1991).

Second, the notion of threat highlights JI as a future-focused phenomenon. JI reflects a forecast about an event, specifically a loss event, which might happen at some point in the future. Thus, not all anticipated or potential job-related events create JI—rather, only those that involve the "potential for harm or loss" (Folkman & Lazarus, 1985: 152; Boswell, Olson-Buchanan, & Harris, 2014). Because threats have not yet materialized, JI involves uncertainty (De Witte, 1999; Probst, 2003; Sverke et al., 2002). Hence, the study of JI is the study of how people perceive and respond to "visualized [job or job feature] loss" (Greenhalgh & Rosenblatt, 1984: 441) as opposed to actual job or job feature loss. Finally, what is under threat in the case of JI is the stability and continuity of one's current employment—specifically, one's current job in one's current organization (De Witte, 1999; Probst, 2003). This differentiates JI from related constructs, such as employability, which captures an individual's perceived ability to obtain a new job, for example, due to "know-how, skills, knowledge of the labour market, and adaptability" (De Cuyper, Bernhard-Oettel, Berntson, De Witte, & Alarco, 2008: 490).

Cleaning Up the Construct

Although job security was part of early models of work motivation, inquiry into JI began in earnest following the widespread layoff and downsizing events that began in the late 1980s. As a result, much of the initial literature was aimed at understanding how employees responded to these types of events (e.g., Jacobson & Hartley, 1991). This initial lens resulted in several features becoming embedded in conceptualizations of JI, including the involuntary nature of threats, the notion of JI occurring when previously secure jobs become insecure, and perceptions of control. These elements are purposely excluded from the current definition in an effort to clarify the JI construct and stimulate research on how these conditions shape responses to threats.

In particular, assuming that JI occurs only in the face of involuntary threats ignores that employees might enter potentially precarious job situations voluntarily (e.g., choosing to work in organizations with selective retention systems or in temporary or contract positions; De Cuyper, De Jong, et al., 2008; Probst, 2003). Conceptualizing JI as occurring only when previously secure work is threatened is likewise problematic because it would suggest that JI research exclude individuals working in more inherently insecure positions (e.g., temporary employees). Finally, considering JI to occur *only* to the extent that people feel powerless to counteract threats confounds the perception of threat with control (Probst, 2003). Importantly, not all threats may be perceived as uncontrollable. For instance, early-stage tenure-track faculty would be expected to experience greater control over their job security than employees facing mass layoffs. These variables are viewed here as moderating conditions, rather than construct assumptions, and are integrated as such into the framework presented in Figure 1.

Future work in JI also requires clarification with regard to whether JI is a cognitive or affective construct. This ambiguity might stem from the fact that the term *insecurity* can refer to cognitive or emotional experiences. The cognitive approach is advocated here given that a

critical question for the JI literature involves understanding the conditions under which job-insecure employees experience anxiety and worry about this threat (Probst, 2003). These reactions—termed affective JI (Huang, Niu, Lee, & Ashford, 2012), job security satisfaction (Probst, 2003), and job loss strain (Ito & Brotheridge, 2007)—thus arguably better reflect proximal outcomes of JI, rather than part of JI itself (Probst, 2003).

In summary, job-insecure employees perceive threats to their jobs or job features. This conceptualization distinguishes JI as a phenomenon that is perceptual, future oriented, and uncertain, and it can refer to potential loss of the job as a whole, as well as deterioration in job conditions and loss of anticipated opportunities. This definition eliminates several assumptions and constraints in prior definitions, thereby inviting such questions as follows: Do individuals respond differently to JI when insecure work is voluntary? How do perceptions of being able to protect one's job—for instance, via high performance—affect reactions to JI? Answering these and other questions requires precise, unconfounded measures of JI.

Implications for Measurement

Table 2 displays a number of commonly used measures of JI. The table is not exhaustive, as studies frequently shorten or adapt measures or use ad hoc items. This practice creates challenges for the comparability of findings across studies—challenges greatly exacerbated by issues of construct contamination in existing measures. The initial columns in Table 2 categorize the measures as capturing quantitative or qualitative JI and illustrate potential construct contamination with affective reactions and perceived control. For instance, the Johnson, Messe, and Crano (1984) scale includes items involving worry over the consequences of job loss, as well as the ability to avoid job loss by working hard. O'Neill and Sevastos' (2013) job loss insecurity scale likewise asks about the respondent's perceived ability to counteract threats. The Ashford et al. (1989) measure includes perceived threats to the job, threats to job features, powerlessness, and importance of job and job features; however, only threats to the job and job features reflect JI as it is currently conceptualized (with the exception of items about moving to "higher" positions). While most common, affective reactions and control are not the only sources of potential construct contamination. For example, the O'Neill and Sevastos (2013: 340) marginalization and organizational survival dimensions were developed inductively from responses to the phrase "I feel insecure in my job when" and seem to capture antecedents rather than JI.

The inclusion of items tapping different constructs makes it difficult to tease out which components are driving effects. In some cases, different components might even have countervailing relationships with other variables. For example, JI and job loss worry exhibit countervailing cross-sectional correlations with organizational commitment (Ito & Brotheridge, 2007; Staufenbiel & König, 2011). While JI threatens commitment, those most committed are most distressed by the prospects of losing their jobs. Also highly problematic is the continued use of multiplicative scoring, which is a feature of the Ashford et al. (1989) measure (e.g., Lee, Bobko, & Chen, 2006). This measure is scored based on the following formula:

$$\textit{Job Insecurity} = [(\Sigma \text{ job feature importance} \times \text{perceived threat to features}) + (\Sigma \text{ importance of total job} \times \text{perceived threat to total job})] \times \text{powerlessness}$$

Table 2
Sample Measures of Job Insecurity

Reference	Sample Items	Scale Captures . . . [a]						
		Qnt Ins	Ql Ins	Aff Rct	Control	Person	Uncert	Time
	Multi-item measures							
Caplan, Cobb, French, Van Harrison, & Pinneau, (1975)	• How certain are you about what your future career picture looks like? • How certain are you about whether your job skills will be of use and value five years from now? • How certain are you about what your responsibilities will be six months from now?		×			×	×	×
Johnson, Messe, & Crano (1984)	• The thought of getting fired really scares me. • I am worried about the disgrace of being fired. • If I get fired, I will not know how to tell people. • If I do good work, my job would be safe. • Working hard would keep me from getting fired.	×		×	×			
Ashford, Lee, & Bobko (1989)	*JI* = [(Σ job feature importance × perceived threat to features) + [(Σ importance of total job × perceived threat to total job)] × powerlessness *Job Feature Importance*: In your worklife, how important are each of the following features to you personally? • Geographic location • Having promotion opportunities • A job that has a significant impact on others. *Perceived Threat to Features*: Looking to the future, what is the probability that changes could occur—changes you don't want or might disagree with that would negatively affect each of these features? *Total Job Importance*: Assume for a moment that each of the following events could happen to you; how important to you personally is the possibility that: • You may lose your job and be moved to a lower level within the organization • Your department or division's future may be uncertain. • You may be moved to a different job at a higher position in your current location. *Perceived Threat to Total Job*: Again, thinking about the future, how likely is it that each of these events might actually occur to you in your current job? *Powerlessness*: • I have enough power in this organization to control events that might affect my job.	×	×		×			
Hellgren, Sverke, & Isaksson (1999)	*Quantitative JI*: • I am worried about having to leave my job before I would like to. • There is a risk that I will have to leave my present job in the year to come. • I feel uneasy about losing my job in the near future.	×		×				
Hellgren et al. (1999)	*Qualitative JI*: • My future career opportunities in [organization] are favourable. • I feel that [organization] can provide me with a stimulating job content in the near future. • I believe that [organization] will need my competence also in the future.		×			×		

(continued)

Table 2 (continued)

Reference	Sample Items	Scale Captures . . . [a]						
		Qnt Ins	Ql Ins	Aff Rct	Control	Person	Uncert	Time
De Witte (2000; see Vander Elst, De Witte, & De Cuyper, 2014)	*JI Scale* • Chances are, I will soon lose my job. • I feel insecure about the future of my job. • I think I might lose my job in the near future. • I am sure I can keep my job.	×						
Probst (2003)	*Job Security Index*: What is the future of your job with this organization like? • sure • uncertain • can depend on being here • up in the air • stable • questionable	×					×	
Probst (2003)	*Job Security Satisfaction Scale*: What is your job security like? • satisfactory • looks optimistic • more secure than most in my job or profession • acceptable	×		×				
Huang, Niu, Lee, & Ashford (2012)	*Affective JI Scale*: • I lose sleep worrying about my future with this company. • I am tense about maintaining my current job employment status. • I am scared by the thought of losing my job. • I am troubled by the thought of losing my job.	×		×				
O'Neill & Sevastos (2013)	*Job Loss Insecurity*: • I am certain of losing my job. • I am uncertain about my future with this organization. • No matter how hard I work there is no guarantee that I am going to keep my job. *Job Changes Insecurity*: • I am expecting unfavorable changes to my job. • I will probably lose many features of my job that I value the most. *Organizational Survival Insecurity*: • Senior management is really trying to build this organization and make it successful. *Marginalization Insecurity*: • I feel as though management is avoiding me.	×	×		×			
Van den Broeck et al. (2014)	*Qualitative JI*: • I feel insecure about the characteristics and conditions of my job in the future. • Chances are, my job will change in a negative way.		×					
	Single-item measures							
De Witte (1999)	How large, in your opinion, is the probability that you will become unemployed in the near future?	×						
Roskies, Louis-Guerin, & Fournier (1993)	What is the probability (0-100) of losing your job within the next year?	×						×

Note: Scale and subscale titles are noted in italics. JI = job insecurity.

[a]Qnt Ins = quantitative insecurity; Ql Ins = qualitative insecurity; Aff Rct = affective reactions; Person = person (vs. job) referent; Uncert = uncertainty (vs. likelihood); Time = time frame.

Multiplicative scales have been highly criticized on conceptual and statistical grounds (Evans, 1991). Specifically, (a) only the highest and lowest possible scores yield unique solutions, making other scores impossible to interpret; (b) not every total score is possible, making the measure ordinal, not interval; and (c) analyses with a multiplicative measure amount to including a cross-product in a regression model without including its constituent components.

A Measurement Agenda

Moving forward, researchers should use scales and items that more precisely assess JI, and they should abandon multiplicative scoring. With this regard, the Probst (2003) and De Witte (2000; see Vander Elst, De Witte, & De Cuyper, 2014) measures focus on overall perceptions of threat of job loss. The latter has been validated in several languages. Research has found measures of overall perceptions to be preferable to composite scales (e.g., Crossley, Bennett, Jex, & Burnfield, 2007), and these measures might outperform measures that ask about specific threats (e.g., total job scale of Ashford et al., 1989), although this remains an empirical question.

Several other measurement issues are important for future research to address, as illustrated in the right three columns of Table 2. First, scrutiny of the items in qualitative JI measures raises questions about the *referents* of some of the scales. The Van den Broeck et al. (2014) and O'Neill and Sevastos (2013) job changes insecurity scales focus on changes to the job and more explicitly capture the notion of potential loss that underlies JI. In contrast, the Caplan, Cobb, French, Van Harrison, and Pinneau (1975) and, perhaps to a lesser extent, the Hellgren et al. (1999) scales appear more oriented to future career prospects and occupational insecurity. Research is needed to examine the extent to which these scales overlap. Research should also investigate whether a person concerned about the need for his or her skills might anticipate job loss; thus, these scales blur quantitative and qualitative JI.

The second issue concerns *uncertainty versus likelihood phrasing* of items. Conceptually, uncertainty is maximized at the midpoint in likelihood scales, where higher scores reflect relatively greater, albeit still imperfect, certainty of loss (i.e., "I am likely to lose my job"). Research is needed to understand how participants respond to these different phrasings and the implications for interpreting results. A third issue concerns the *time frame* of scaling. Roskies and Louis-Guerin (1990) suggest that bounding JI within a particular time frame (e.g., probability of losing one's job in the next year) might help distinguish acute and chronic JI, although it is important to note that this cannot capture variability in JI over time. However, Manski and Straub (2000) cautioned that because people might quit to preempt job loss, intentions to quit might confound items inquiring about likelihood of loss in a particular time frame. Future research should explore this potential confound and whether it is helpful to instruct participants to answer items independently of intentions to quit.

In summary, JI research needs to use uncontaminated measures that facilitate clear interpretations of results. Several existing measures appear to meet these criteria. Yet, research is needed to examine questions related to (a) global versus facet measures, (b) person versus job referents in qualitative JI scales, (c) uncertainty versus likelihood phrasing, and (d) whether intentions to quit serve as a confound for measures of JI.

Table 3
Summary of Research on Antecedents of Job Insecurity

Antecedents	Theorized Relationship	Theorized Mechanisms	Illustrative References and Empirical Findings[a]
		National/macroeconomic	
Poor economic condition (e.g., unemployment rate)[b]	↑	Stimulate organizational efforts to reduce costs; distal warning sign of a potential threat.	Anderson & Pontusson (2007): $b = .03$ ($SE = .02$), controlling for policy, change in economic condition, demographics. Lübke & Erlinghagen (2014): $b = .16$ ($SE = .04$), controlling for demographics.
Decline in economic condition (e.g., change in unemployment rate)[b]	↑	See above; individuals and organizations are particularly attuned to changes in their environments.	Anderson & Pontusson (2007): $b = .09$ ($SE = .04$), controlling for policy, economic condition, demographics. Lübke & Erlinghagen (2014): $b = .12$ ($SE = .03$), controlling for economic condition and demographics.
Industry decline, shrinking demand[b]	↑	Stimulate job-threatening changes as organizations compete to survive; distal warning sign of potential threats.	Roskies & Louis-Guerin (1990): perceived changes in business or technological environment predicted insecurity about termination and demotions and long-term job insecurity ($\beta = .16, .13$), controlling for personal and environmental factors. Jiang, Probst, & Sinclair (2013): occupational outlook predicted individual job insecurity ($b = -.02$), controlling for demographics.
Technological change[b]	↑	See above.	See above.
Employment protection legislation	↓	Objectively makes jobs more secure by placing restrictions on firing workers.	Anderson & Pontusson (2007): $b = -.20$ ($SE = .10$), controlling for economic conditions and demographics.
		Organizational	
Company performance	↓	Poor performance is often a precursor to downsizing.	Debus, König, & Kleinmann (2014): $r = -.28$, company-level correlation.
Changes in leadership	↑	Considered a proximal warning sign of potential changes to one's job.	Roskies & Louis-Guerin (1990): composite of management changes, formal announcements, and rumors predicted insecurity about working conditions, insecurity about termination and demotions, and long-term job insecurity ($\beta = .24, .24, .17$), controlling for personal and environmental factors.
Organizational change	↑	Objective threat; provides environmental signals that job is at risk.	Keim, Landis, Pierce, & Earnest (2014): meta-analytic $r_c = .18$.
Organizational communication	↓	Poor communication enables rumors, creates unclear expectations.	Keim et al. (2014): meta-analytic $r_c = -.24$.
Fairness of past change	↓	Expect fairness to continue and to limit risk.	Ito & Brotheridge (2007): $r = -.31$.
Use of temporary or contingent contracts	↑	Signal that the organization is less committed to long-term employment; temporary workers may compete with permanent workers.	De Cuyper, Sora, et al. (2009): r(Belgium) = .70, r(Spain) = .37, between percentage of temporary workers and job insecurity climate.
Union presence	↓	Provides institutional protection for jobs.	Anderson & Pontusson (2007): union membership ($b = -.15$, $SE = .03$), controlling for policy, economic conditions, and demographics.
Sector (private vs. public)	↓	Higher objective risk in private sector; public sector generally perceived as safe.	Anderson & Pontusson (2007): $b = -.31$ ($SE = .03$), controlling for policy, economic conditions, and demographics.
Employee involvement	↓	Reduces uncertainty.	Huang, Niu, Lee, & Ashford (2012): $r = -.18$
		Positional	
Formal contract (temporary, contingent vs. permanent)	↑	Objective threat as part of the contract; lower attachment to the organization and more peripheral work.	Keim et al. (2014): meta-analytic $r_c = .29$. Klandermans, Hesselink, & Van Vuuren (2010): Compared with employees with other contracts, agency workers (D = .12) and those with a fixed contract (D = .08) experienced greater job insecurity.
Part-time employment status	↑	Objective threat; more peripheral position may create perception of insecurity.	Keim et al. (2014): meta-analytic $r_c = .05$, *ns*. Debus et al. (2014): $r = .03$, *ns*. See also arguments in Näswall & De Witte (2003).
Manual/blue-collar work[b]	↑	Lower levels of human capital; work in industries more prone to layoffs.	Keim et al. (2014): meta-analytic $r_c = .19$. Anderson & Pontusson (2007): $b = .09$ ($SE = .03$), controlling for policy, economic conditions, and demographics.

(continued)

Table 3 (continued)

Antecedents	Theorized Relationship	Theorized Mechanisms	Illustrative References and Empirical Findings[a]
		Individual: Demographic	
Organizational tenure	↓	Higher tenure is associated with more firm-specific skills.	Lübke & Erlinghagen (2014): $b = –.04$ ($SE = .01$), controlling for demographics.
Age	↑/↓	Older workers could have a perceptual sensitivity to threats; in contrast, employers take seniority into account in reductions.	Keim et al. (2014): meta-analytic $r_c = –.07$. Anderson & Pontusson (2007): $b = –.01$ ($SE = .00$), controlling for policy, economic conditions, and demographics. Dekker (2010): $\beta = –.08$, controlling for demographics.
Women[b]	↑/?	Lower power in labor market creates sensitivity to threats; employment in more precarious jobs, but demand and discrimination protection might reduce threats.	Keim et al. (2014): meta-analytic $r_c = –.03$, *ns*. Anderson & Pontusson (2007): $b = –.04$ ($SE = .02$), *ns*, controlling for policy, economic conditions, and demographics. Dekker (2010): $\beta = –.07$, controlling for demographics. Lübke & Erlinghagen (2014): $b = .15$ ($SE = .04$), controlling for demographics.
Minority status[b]	↑	Discrimination; disparities in occupational characteristics and opportunities for human capital development; perceptual bias based on negative workplace experiences.	Yang & Zheng (2015): odds for black workers experiencing job insecurity as compared with white workers = 2.47, controlling for human capital and organizational factors.
Education level[b]	↓	Education is a key component of human capital and therefore promotes job security.	Keim et al. (2014): meta-analytic $r_c = –.04$, *ns*. Anderson & Pontusson (2007): $b = –.02$ ($SE = .00$), controlling for policy, economic conditions, and demographics.
		Individual: Experiential	
Perceived employability[b]	↓	Employers use job security to attract and retain employable workers; employable less likely to appraise events as job threatening.	De Cuyper, Bernhard-Oettel, Berntson, De Witte, & Alarco (2008): $r = –.14$. De Cuyper, Mäkikangas, Kinnunen, Mauno, & De Witte (2012): $r = –.18$ with job insecurity assessed 1 year later, cross-lagged panel coefficient over 1 year = –.08.
Training experiences	↓	Investment in human capital.	Kohlrausch & Rasner (2014): $b = –.20$ ($SE = .09$), relationship strengthens for those with lower levels of education.
Role ambiguity, role conflict	↑	Uncertainty about role and performance leads to perception of job at risk.	Keim et al. (2014): meta-analytic r_c ambiguity = .28, conflict = .20.
Emotional exhaustion	↑	Exhausted employees might be vulnerable to job loss due to poorer performance.	De Cuyper, Mäkikangas, Kinnunen, Mauno, & De Witte (2012): $r = .16$, with job insecurity assessed 1 year later, cross-lagged panel coefficient = .05.
Prior unemployment	↑	Creates predisposition to perceive threats; psychological spillover.	Ellonen & Nätti (2015; 2008 model): logistic coefficient = 1.2 predicting dichotomized job insecurity. Roskies & Louis-Guerin (1990): predicted working condition insecurity, insecurity about termination and demotion, and long-term insecurity ($\beta = .15, .18, .09$), controlling for personal and environmental factors.
		Individual: Personality	
External locus of control[b]	↑	Predisposition to view one's future as controlled by environmental events.	Keim et al. (2014): meta-analytic $r_c = .25$. Debus et al. (2014): $r = .38$.
Negative affectivity[b]	↑	Tendency to appraise events negatively.	Debus et al. (2014): $r = .19$.
Low self-esteem[b]	↑	Tendency to perceive self as unworthy and incapable.	Kinnunen, Feldt, & Mauno (2003): self-esteem cross-lagged panel coefficient over 1 year, $\beta = –.08$.
Low core self-evaluations[b]	↑	Tendency to perceive threats in one's environment.	Låstad, Berntson, Näswall, & Sverke (2014): core self-evaluations associated with quantitative and qualitative job insecurity ($r = –.27, –.29$) assessed 1 year later.
		Interpersonal	
Workplace bullying	↑	Bullying can involve threatening jobs and creating undesirable job conditions.	Glambek, Matthiesen, Hetland, & Einarsen (2014): $r = .24$ with job insecurity assessed 6 months later.

[a]Corrected meta-analytic correlations (r_c) are provided for those predictors included in the Keim et al. (2014) meta-analysis. Additional illustrative references are not included in the meta-analysis. Preference is given to large-scale studies (e.g., Anderson & Pontusson, 2007) in listing illustrative references. All findings are statistically significant unless followed by *ns*. Bivariate correlations are presented if provided in the original paper.

[b]Antecedent conditions that enhance both job insecurity and vulnerabilities, thus creating particularly perilous conditions.

Table 4
Typology of Mechanisms and Foci Linking Antecedents to Job Insecurity

	Mechanism	
Threat Focus	Selection	Subjective Perception
Job at risk	Employed in at-risk jobs (e.g., temporary work)	Heightened tendency to perceive greater risk to the job
Person at risk	Create circumstances that put self at risk (e.g., poor performance, poor relationship with supervisor)	Heightened tendency to perceive risk to self as jobholder

Sources of JI

The majority of research has focused on JI's consequences, with relatively less research devoted to antecedents. To guide future work, this section reviews past research on antecedents, suggests potential mechanisms linking antecedents to JI, and proposes several new categories of predictors. To lay the groundwork for this discussion, Table 3 summarizes the current state of research regarding antecedents theorized to affect JI.

Antecedents

Although JI is a perceptual phenomenon, these perceptions generally have their basis in environmental threats (Klandermans, Hesselink, & Van Vuuren, 2010). For instance, JI tends to rise with a rising national unemployment rate (Anderson & Pontusson, 2007) and in response to technological and other changes in the business environment (e.g., globalization) that contribute to industry and occupational decline, shrinking demand, and the outsourcing of labor (Jiang, Probst, & Sinclair, 2013; Lübke & Erlinghagen, 2014; Roskies & Louis-Guerin, 1990). From a policy perspective, employment protection legislation (e.g., notice periods, severance pay) enhances the difficulty of firing workers and reduces JI (Anderson & Pontusson, 2007).

At the company level, company performance, management changes, as well as formal and informal announcements of impending changes serve as more proximal warning signs that one's job might be at risk and are accordingly associated with increased JI (Debus, König, & Kleinmann, 2014; Ellonen & Nätti 2015; Keim, Landis, Pierce, & Earnest, 2014; Roskies & Louis-Guerrin, 1990). Union membership reduces JI, while private sector employment increases it (Anderson & Pontusson, 2007). Consistent with the notion that temporary positions are inherently insecure, temporary employees report greater JI (Keim et al., 2014). Interestingly, permanent workers in organizations that employ greater proportions of temporary workers report greater JI (De Cuyper, Sora, De Witte, Caballer, & Peiró, 2009). Finally, consistent with economic and technological trends that threaten blue-collar work, blue-collar workers report higher levels of JI (Kalleberg, 2011; Keim et al., 2014).

Individual variables are also relevant. Debus et al. (2014) found that the personality traits of negative affectivity and locus of control explained nearly double the amount of variance in JI perceptions as did company performance and contract type. Other traits, such as low core self-evaluations and low self-esteem, as well as several experiential (e.g., emotional

exhaustion) and demographic factors (e.g., minority status, tenure), have been linked to JI (De Cuyper, Mäkikangas, Kinnunen, Mauno, & De Witte, 2012; Låstad, Berntson, Näswall, & Sverke, 2014; Yang & Zheng, 2015). Although one might expect to find person-environment interactions, the existing evidence is somewhat weak (Debus et al., 2014). The general logic is that environmental conditions create threats, while individual characteristics create heightened vigilance to these threats. However, research is needed to examine the extent to which various antecedent conditions indeed operate in such a manner.

Mechanisms and Threat Foci

It remains unclear if various antecedents, especially those at the individual level, engender JI because they are objectively associated with increased risk and/or because they create a perceptual bias leading to greater risk perception. For example, is prior unemployment associated with JI (Ellonen & Nätti, 2015) because it leads one to accept more objectively insecure work (e.g., temporary or contingent work) or because it creates a "psychological spillover" (P. R. Davis, Trevor, & Feng, 2015) that predisposes a person to perceive greater threats? The former is indicative of a selection mechanism, whereby antecedents lead to JI by virtue of creating more objective threats—for example, by stimulating employment in at-risk jobs or behavior that puts one's job at risk. In contrast, perception mechanisms capture that antecedents can influence how individuals subjectively perceive their environment by (a) stimulating higher baseline perceptions of threats and/or (b) amplifying or lessening JI in response to more objective threats. Empirically, selection mechanisms raise the possibility that some antecedents affect JI indirectly. For example, if prior unemployment predicts JI because it leads people to be disproportionately employed in more insecure jobs (e.g., temporary positions), one would expect to find that contract type mediates the prior unemployment-JI relationship. However, one would expect (a) direct effects if prior unemployment creates higher overall levels of threat perception and (b) moderated effects if prior unemployment serves to amplify JI in response to temporary contracts or other threats.

To provide greater insight into these mechanisms, Table 4 introduces a 2 × 2 typology that crosses the selection/perception mechanisms with the specific foci of threats. Although surprisingly ignored in the literature, Jacobson and Hartley (1991) noted that threats to job security can be person independent, meaning that the job is at risk regardless of the holder (labeled here *job-at-risk threats*), or person dependent, meaning that one's continued employment in a given job is at risk (*person-at-risk threats*) although the position itself may continue. This typology suggests that antecedents give rise to JI by virtue of selection and/or perception mechanisms related to job-at-risk or person-at-risk threats.

Uncovering New Predictors

The typology is valuable not only for understanding how and why antecedents relate to JI but also for expanding research on antecedents of JI. This is especially the case for person-at-risk predictors, which have been largely neglected in the JI literature. To this aim, five new sets of antecedents are ripe for empirical investigation: individual performance, organizational performance management systems, organizational culture, supervisor-subordinate relationships, and self-employment. For example, poor task performance, high absenteeism,

and counterproductive work behaviors, although typically viewed as outcomes of JI, should predict JI by virtue of placing individual jobholders at risk. People working in organizations using forced ranking systems (especially if rankings are tied to reductions in force), in organizations with competitive cultures, and in highly political environments likely experience greater JI. These environments might not only create threats for particular jobholders but also enhance vigilance to potential threats. Poor relationships with supervisors might induce threats via selection and/or perceptual mechanisms (Hershcovis & Barling, 2010). Indeed, participants in O'Neill and Sevastos's (2013) qualitative study reported that that they felt insecure about the future of their jobs when they perceived that superiors were avoiding them and when they felt excluded from relevant workplace conversations. In contrast, high-quality relationships with a supervisor might shield employees from risk (Lawrence, Halbesleben, & Paustian-Underdahl, 2013). Finally, JI research has largely ignored the self-employed, although Schonfeld and Mazzola's (2015) qualitative study revealed several sources of JI among self-employed workers. Importantly, for self-employed workers, the line between job-at-risk threats and person-at-risk threats may be blurred. Clearly, much more research is needed to fully map the antecedent space.

Summary

Environmental conditions as well as individual factors predict JI. They do so by creating threats and/or stimulating enhanced perception of threats. The position itself (regardless of holder) might be threatened, or threats might be localized to specific individuals. The proposed 2 × 2 typology of mechanisms and threat foci aims to stimulate (a) greater insight into why and how antecedent conditions predict JI and (b) the identification of additional antecedents, including personal performance, management strategies, and supervisor relationship quality.

Reactions to JI: Mechanisms and Outcomes

As depicted in Figure 1, JI relates to outcomes via four overarching mediating mechanisms: stress-related mechanisms, social exchange–related mechanisms, job preservation motivation, and proactive coping. The following sections describe the mechanisms and consider corresponding and competing predictions made among them.

Stress-Related Mechanisms

Several rationales position JI as a stressor (De Witte et al., 2016). First, JI threatens "vitamins" (Warr, 1987) and resources that are key to well-being and acquired through work, including identity, income, social connection, and social status (Jiang & Probst, 2014; Schreurs, Van Emmerik, Notelaers, & De Witte, 2010). In this vein, characterizations of JI as being stressful have drawn from work on unemployment (e.g., Jahoda, 1982) to illustrate latent (e.g., social value) and manifest (e.g., income) benefits of work threatened by JI (De Witte, 1999; Sverke et al., 2002; Vander Elst, Näswall, Bernhard-Oettel, De Witte, & Sverke, 2016). Second, researchers point out that uncertainty itself is stressful (De Witte, 1999). Third, JI violates basic psychological needs for autonomy, competence, and relatedness (Deci & Ryan, 2000; Vander Elst, Van den Broeck, De Witte, & De Cuyper, 2012; Van den

Broeck et al., 2014). These rationales align well with broader appraisal and resource-based models of stress (e.g., Folkman & Lazarus, 1985; Hobfoll, 1989). Consistent with the notion that anxiety and worry would be expected to result from the appraisal of valued resources as being threatened, the measures of affective responses to JI in Table 2 could be used as indicators of stress-related responses. While some view a lack of situational control as a mediator of JI effects (e.g., Vander Elst, De Cuyper, Baillien, Niesen, & De Witte, 2016), control over threats (i.e., situational control) is treated as a moderator here given that not all threats might be viewed as uncontrollable.

These explanations point to JI as a formidable stressor that results in a variety of strain outcomes, including poor well-being, diminished job attitudes, and poor performance, which the evidence largely supports (Cheng & Chan, 2008). This is especially the case with well-being, where the effects of JI mirror the effects of actual loss (De Witte, 1999). Moreover, given that stress narrows attention and diminishes self-regulation, stress mechanisms predict that JI results in increased workplace injuries and accidents, along with diminished creativity and increased instances of interpersonal mistreatment and organizational counterproductive work behavior (De Cuyper, Baillien, & De Witte, 2009; Niesen, De Witte, & Battistelli, 2014; Probst, 2004; Van den Broeck et al., 2014). These outcomes are discussed later, as they are the subject of corresponding and competing predictions made by several perspectives. That JI threatens key purposes and benefits of working and frustrates basic psychological needs predicts outcomes beyond the workplace. For example, several studies find that parents' JI is negatively associated with children's academic performance, career self-efficacy, work beliefs, and work attitudes (e.g., Barling, Dupré, & Hepburn, 1998; Lim & Sng, 2006; Zhao, Lim, & Teo, 2012). Arguments in the group conflict literature suggest that JI might worsen attitudes toward groups seen as competing for jobs. Indeed, JI is weakly associated with negative attitudes toward immigration (Billiet, Meuleman, & De Witte, 2014). Research is needed to investigate potential consequences for the workplace, such as discrimination.

Social Exchange–Related Mechanisms

Social exchange theories offer another lens through which to view JI—namely, as an imbalance in the exchange relationship between employee and employer. From the perspective of psychological contract theory, JI reflects a breach of the relational psychological contract, wherein employees exchange loyalty and commitment for security and other socioemotional benefits (De Cuyper & De Witte, 2006, 2007). Viewed through the lens of justice theory, JI lessens the rewards that employees receive for their efforts and is therefore likely to be perceived as unjust (Piccoli & De Witte, 2015). Thus, the former suggests that employees view JI as an imbalance between obligations and inducements; the latter, between inputs and outcomes.

Like the stress perspective, social exchange mechanisms (e.g., Gakovic & Tetrick, 2003; Siegrist, 1996) predict that employees will respond to JI with ill-being, lessened job attitudes, and lessened contributions to the organization (De Cuyper & De Witte, 2006; Piccoli & De Witte, 2015; Reisel, Probst, Chia, Maloles, & König, 2010). JI can also violate exchanges beyond the employee-organization relationship. Employees view JI as a breach of the psychological contract between union members and unions, leading to intentions to resign

membership (De Witte et al., 2008), as well as between individuals and their elected political officials, leading to diminished trust in politicians and political institutions (Wroe, 2014).

Job Preservation Motivation

Because threats have not yet manifested, JI might motivate employees to act in ways that they believe might keep job or job feature loss from occurring (i.e., job preservation motivation). Job preservation strategies can take several forms. For example, employees might endeavor to demonstrate their worth to the employer by devoting extra effort toward behaviors that will be noticed and valued. For example, research finds that job-insecure employees enact more safety behaviors when the organization values safety (Probst, 2004). This line of reasoning predicts that JI results in enhanced task performance and organizational citizenship behavior, as these behaviors are likely to be noticed and valued by higher-ups (Shoss & Probst, 2012). Other active job preservation efforts include attempting to portray oneself as an ideal employee, for example by "suppressing personal values and pretending to embrace organizational values" (i.e., facades of conformity; Hewlin, Kim, & Song, 2016: 539) and being willing to make concessions (e.g., lower pay, longer hours; Otto, Hoffman-Biencourt, & Mohr, 2010). Thus, JI might result in work intensification as a job preservation strategy on the part of employees.

Employees might also pursue more passive strategies in an attempt to "fly under the radar" and refrain from behavior that would draw attention to them and put their jobs at greater risk. Evidence suggests that job-insecure employees avoid alerting the organization to potential problems (Schreurs, Guenter, Jawahar, & De Cuyper, 2015), engage in presenteeism (Miraglia & Johns, 2016), and are less likely to take advantage of work-nonwork support programs (Boswell et al., 2014). This logic predicts that job-insecure employees enact fewer organizationally directed counterproductive behaviors (Probst et al., 2007). However, given that managers are often inattentive to interpersonal behaviors during times of uncertainty, job preservation motivation predicts that employees might mistreat coworkers as a means to compete with potential rivals (Shoss & Probst, 2012). Research is needed to identify strategies that people might use to compete with others, such as hoarding knowledge (S. Wang & Noe, 2005).

Future work should consider that job preservation motivation might not only translate into efforts to demonstrate one's own worth as an employee, but also translate into efforts to demonstrate the value of the particular position. For example, job-insecure employees might try to convince higher-ups of the importance of the job or take steps to increase the job's importance (i.e., job aggrandizement). They might also pursue collective strategies—for example, by joining unions (Bender & Sloane, 1999) or engaging in protests (Klandermans, Van Vuuren, & Jacobson, 1991). That said, this is a complex issue (see Sverke et al., 2004), and some argue that individuals might avoid collective action for fear that doing so will threaten their jobs (Kalleberg, 2011).

Proactive Coping

Job preservation motivation captures individuals' attempts to avoid loss. In contrast, proactive coping captures people's attempts to cope ahead of time with job or job feature loss, although loss is uncertain (see Aspinwall & Taylor, 1997). For instance, some might network or proactively seek out other jobs to preempt job loss or gauge the market (Klehe, Zikic, Van

Vianen, Koen, & Buyken, 2012). Job-insecure persons might also pursue educational opportunities to enhance potential job opportunities, which future research should investigate. They might increase savings to buffer against potential income loss. Although relatively unexplored, research does find that job-insecure employees report withholding purchases of consumption goods (Lozza, Libreri, & Bosio, 2012). JI also features prominently in demand-driven explanations for social insurance, as those who perceive themselves to be at risk seek greater protection in case of loss (Dekker, 2010).

Corresponding and Competing Predictions

As illustrated in a general manner in Figure 1, the four mechanisms make several corresponding as well as competing predictions regarding outcomes. The paragraphs that follow describe corresponding and conflicting predictions and associated empirical evidence.

Corresponding predictions. *Stress*- and *social exchange*–related mechanisms predict that JI undermines personal and work-related well-being as well as job attitudes and performance, albeit for different reasons (i.e., strain and diminished self-regulation and attention effects vs. an imbalance in the social exchange relationship; Niessen & Jimmieson, 2016; Reisel et al., 2010; Sverke & Hellgren, 2002). In accordance with these arguments, longitudinal work links JI to poor mental and physical health (De Witte et al., 2016), burnout (De Cuyper et al., 2012), and diminished self-esteem (Kinnunen, Feldt, & Mauno, 2003). Job-insecure employees also report lower levels of job satisfaction, organizational commitment, and trust (Cheng & Chan, 2008). Cross-sectional evidence suggests that social exchange–related mechanisms, especially injustice, might be a stronger mediator of well-being and job attitudinal effects (Piccoli & De Witte, 2015; Vander Elst, De Cuyper, et al., 2016). However, longitudinal research is needed to truly assess mediation and to compare the relative strength of these mechanisms across time. Performance is a subject of competing predictions and is therefore discussed below.

While the stress and social exchange perspectives are most closely aligned, other perspectives do make similar predictions regarding specific outcomes. *Stress* and *job preservation motivation* both predict poor interpersonal outcomes, and indeed JI has been linked to bullying, diminished coworker support, and diminished coworker satisfaction (De Cuyper et al., 2009; Ferrie et al., 2001; Probst, 2005). From the stress perspective, this occurs as a result of displaced aggression and diminished self-regulation. From the perspective of job preservation motivation, these are instrumental behaviors designed to compete with potential rivals (Shoss & Probst, 2012). *Stress, social exchange*, and *proactive coping* all predict that JI reduces intentions to remain, either as a response to a stressful environment, a response to a damaged social exchange relationship, or to preempt loss. The empirical evidence supports this prediction (Cheng & Chan, 2008). Finally, it is worth noting that some *job preservation* strategies might reflect *proactive coping* to the extent that they are done to facilitate coping with potential loss. For instance, employees might try to curry favor with supervisors to receive a positive recommendation or to position for a more secure job within the same organization.

Competing predictions. The *stress* and *social exchange* mechanisms suggest that JI creates stress and damages social exchange relationships, with the consequence of lessening employees' willingness to put forth effort to help the organization and making it difficult

for employees to perform. Yet, *job preservation motivation* suggests that JI might motivate employees to work harder for the instrumental purpose of trying to secure their jobs. Staufenbiel and König (2010) found evidence of these competing tensions in a cross-sectional study. Specifically, they found (a) a weak positive direct effect of JI on supervisor-rated performance and (b) a stronger negative indirect effect through weakened job attitudes, leading to an overall negative effect of JI on performance (see Stynen, Forrier, Sels, & De Witte, 2015). Consistent with this finding, meta-analysis reveals an overall negative relationship between JI and performance (Cheng & Chan, 2008). Stress effects appear to be particularly detrimental to the extent that performance requires creativity and adaptability (Niessen & Jimmieson, 2016; Niesen et al., 2014). However, studies have found moderated and/or curvilinear effects of JI on performance (Brockner et al., 1992; Selenko, Mäkikangas, Mauno, & Kinnunen, 2013) and citizenship behaviors (C. F. Lam, Liang, Ashford, & Lee, 2015), suggesting that job preservation motivation might outweigh stress/social exchange effects at certain levels of JI and depending on the circumstances. Studies finding positive and negative relationships between JI and organization-focused counterproductive work behavior (Probst et al., 2007; Reisel et al., 2010) likewise suggest the existence of moderators.

Another competing prediction comes from the fact that *job preservation* and *proactive coping* both require attention and effort. In many cases, effort devoted to one is unlikely to be available to be devoted to the other. For this reason, job preservation motivation and proactive coping could be viewed as competing responses to JI, where those focused on trying to preserve their jobs are unlikely to devote energy toward proactive coping. In this sense, one might expect *stress* mechanisms to conflict with *proactive coping*, as the negative emotional arousal associated with greater stress-related reactions to JI might make it difficult for employees to devote the energy and attention required to proactively cope (Aspinwall & Taylor, 1997). Finally, while the *social exchange* perspective predicts that employees will see JI as a failure on the part of the union to protect them, the *job preservation* perspective suggests that job-insecure employees might turn to the union for support (Sverke et al., 2004). Future research is needed to examine the nature and timing of these effects.

Summary

The previous sections described four main mechanisms through which JI operates to effect outcomes. These reflect stress- and social exchange–related consequences, as well as strategies that employees pursue to avoid loss and to proactively cope with loss should it occur. They yield important questions for future research: What are the strategies that individuals use to try to preserve their job or job features? What are the ways by which people proactively cope with potential job or job feature loss? How do competing tensions among the mechanisms play out over time? Although a full discussion is omitted for space, research is also needed to explore potential feedback mechanisms. For example, there is some evidence that job preservation strategies can ultimately harm well-being and job attitudes (Boswell et al., 2014; De Cuyper, Schreurs, Vander Elst, Baillien, & De Witte, 2014; Hewlin et al., 2016). However, these strategies could, as intended, reduce JI (De Cuyper et al., 2014). Proactive coping might lessen concerns about being able to find an equal or better job should loss occur (affecting economic vulnerabilities, as detailed next). These questions echo the importance of longitudinal research on JI.

Moderators: Impact and Key Drivers

The previous sections reviewed general patterns of reactions to JI. The following sections describe three sets of moderating factors—threat features, economic vulnerabilities, psychological vulnerabilities—expected to shape reactions to JI and identify potential key drivers of each.

Threat Features

As noted in the construct clarification section, not all experiences of JI are the same, especially in terms of perceived situational control, duration, and volition. While situational control has received some attention, particularly as a conceptual explanation for the effects of variables identified here as key drivers of situational control, the other features have received less research attention and are presented largely as directions for future research.

Perceived situational control. Perceived situational control concerns individuals' appraisals regarding the degree to which a given threat can be counteracted—in other words, whether there are ways to keep the threat of loss from translating into actual loss. In that sense, it can be differentiated from job control, dispositional control, and other conceptualizations (Vander Elst, De Cuyper, et al., 2016). To the extent to which people view threats as controllable, they are more likely to be motivated to take action to try to avert threats. Thus, perceived situational control should enhance job preservation motivation in light of JI (Klandermans et al., 1991). Indeed, employees appear to respond to JI with greater citizenship behaviors when threats are tied to performance (Wong, Wong, Ngo, & Liu, 2005). Stress theory also posits that situational control will reduce stress-related effects (Folkman & Lazarus, 1985).

Threat duration. From a temporal perspective, threats may vary from *acute* to those that are more *chronic* in nature (Roskies, Louis-Guerin, & Fournier, 1993). Prolonged uncertainty and vigilance associated with more chronic threats would be expected to enhance stress-related mechanisms, resulting in a stronger relationship between JI and stress-related outcomes (Burchell, 2011; Ferrie et al., 2001). Indeed, in one of the few studies to address duration, Burgard et al. (2009) found that the detrimental well-being impact of JI compounds over time. As a related consequence, prolonged uncertainty should make it difficult for individuals to sustain effort toward job preservation strategies and proactively cope.

Volition. As previously noted, some might enter insecure job situations voluntarily—for example, because these situations offer flexibility, opportunities for training, or some other benefits (e.g., for a discussion of voluntary temporary workers, see De Cuyper, De Jong, et al., 2008; for a discussion of voluntary self-employed, see Schonfeld & Mazzola, 2015). People might also support decisions in their organizations that enhance JI (e.g., about introducing new technologies). Much remains unknown about how volition shapes reactions to JI. From a theoretical standpoint, personal choice would be expected to assuage the extent to which JI frustrates autonomy needs, thus buffering stress-related effects (De Cuyper, De Jong, et al., 2008; Deci & Ryan, 2000). Moreover, those entering job-insecure situations voluntarily might be more attuned to positive aspects of these situations, with the consequence of reducing negative stress and social exchange consequences of JI (De Cuyper, De Jong,

et al., 2008; De Cuyper & De Witte, 2007). That said, De Cuyper and De Witte (2007) found that JI affects commitment more for voluntary than involuntary temporary workers, but this unexpected finding might be due to the operationalization of volition. Those who are involuntarily in a job-insecure situation would be expected to put forth greater effort in an attempt to make their jobs more secure. However, those who enter these situations voluntarily might view them more positively and respond with greater effort in order to remain. For an excellent discussion of these issues with reference to temporary workers, see De Cuyper, De Jong, et al. (2008).

Key drivers of threat features. Threat features provide levers through which various factors shape reactions to JI. For instance, job control or participative decision making could afford individuals opportunities for situational control (Probst, 2003). Studies have revealed that these variables buffer the negative impact of JI on several outcomes, including satisfaction, withdrawal, engagement, and health complaints (Barling & Kelloway, 1996; Probst, 2005; Schreurs et al., 2010; but see Vander Elst et al., 2011). Organizational communication, organizational justice, and a high-quality relationship with one's supervisor might promote perceptions of situational control by clarifying expectations and signaling that effort will be rewarded. Accordingly, research finds buffering effects of these variables on several negative effects of JI (Jiang & Probst, 2014; Loi, Ngo, Zhang, & Lau, 2011; Schreurs, Van Emmerik, Günter, & Germeys, 2012; Sora, Caballer, Peiró, Silla, & Gracia, 2010; H. Wang, Lu, & Siu, 2015). Although unions have been theorized to enhance the perceived controllability of threats, research is unclear with this regard (Hellgren & Chirumbolo, 2003). The source of JI is also relevant. For example, announcements of impending layoffs would spur a relatively acute experience of JI. Finally, personality- and career-related factors might affect voluntary employment in more precarious jobs (De Cuyper, De Jong, et al., 2008; Schonfeld & Mazzola, 2015).

Economic Vulnerabilities

Economic vulnerabilities refer to concerns about job and income replacement should loss occur (Anderson & Pontusson, 2007; Berglund, Furåker, Vulkan, 2014). Specifically, *labor market insecurity* reflects insecurity about one's prospects of finding a job of equal or better quality on the external labor market. Labor market insecurity corresponds well with the notion of perceived external employability, as articulated by Vanhercke, De Cuyper, Peeters, and De Witte (2014). *Income insecurity* reflects concerns about income replacement should loss occur. The notion of economic vulnerabilities has its roots in earlier discussions about job dependence. However, the meaning of the term *job dependence* is somewhat ambiguous, and it has been used in several ways (e.g., Cheng & Chan, 2008; Greenhalgh & Rosenblatt, 1984; Ito & Brotheridge, 2007; Richter, Näswall, Bernhard-Oettel, & Sverke, 2014).

High levels of economic vulnerabilities heighten the threat posed by JI and heighten stress-related reactions. In accordance with this idea, research finds that those who are concerned about being able to find a new job and/or replace lost income have more negative affective reactions to JI (Anderson & Pontusson, 2007; Berglund et al., 2014; Ito & Brotheridge, 2007). Workers with fewer labor market prospects, such as manual and blue-collar workers, respond to JI with increased strain (Sverke et al., 2002). One might anticipate that those more economically vulnerable are more motivated to try to secure their jobs

because they anticipate greater challenges replacing the job and/or income should loss occur. However, given the heightened stress reactions, it is unclear whether such efforts are likely to succeed. Those most economically vulnerable may have challenges coping proactively because they have less initial resources on which to draw and potentially less human capital (Vanhercke et al., 2014). In support of this idea, JI is more strongly associated with turnover intentions among those who report greater perceived employability (Berntson, Näswall, & Sverke, 2010), suggesting that those who are the most employable have greater opportunities to proactively cope.

Key drivers of economic vulnerabilities. Economic vulnerabilities are linked to context, defined by Johns (2006) in an omnibus sense as reflecting the journalist's heuristics of where (e.g., country, industry), when (e.g., historical time), who (e.g., occupation, demographic contexts), and why (e.g., rationale for data collection). The first three are most relevant here and together create individuals' "situational opportunities and constraints" (Johns, 2006: 386) for replacing loss (see e.g., Vanhercke et al., 2014). For example, research across a number of Organisation for Economic Co-operation and Development countries finds that country-level unemployment benefits are negatively associated with worries over JI (Anderson & Pontusson, 2007). Moreover, country-level labor market (e.g., job search assistance, worker training) and income security (e.g., unemployment benefits) expenditures buffer the impact of JI on job attitudes (Debus, Probst, König, & Kleinmann, 2012).

Also relevant is the state of the economy, as both income and labor market insecurity increase along with a deteriorating state of the economy (Anderson & Pontusson, 2007; Lübke & Erlinghagen, 2014). Accordingly, the negative impact of JI on well-being has been found to be stronger in times of economic downturn (J. Lam, Fan, & Moen, 2014). Jobs that provide learning opportunities have been found to reduce labor market insecurity (Dekker, 2010) and thus would be expected to buffer JI effects. Likewise, competency-based employability (Berntson et al., 2010; Vanhercke et al., 2014), personal and professional networks, as well as traits (e.g., external locus of control; Dekker, 2010) and demographic factors (e.g., education, age; Anderson & Pontusson, 2007; Cheng & Chan, 2008; Manski & Straub, 2000) that affect labor market insecurity would be expected to contribute to greater worry over threats and associated detrimental consequences. Finally, household income and savings and, especially in the United States, access to alternative sources of health care coverage might reduce economic vulnerabilities (Schonfeld & Mazzola, 2015). To summarize, economic vulnerabilities—including both labor market and income insecurity—produce more detrimental reactions to JI. Key drivers include national social safety net policies, macroeconomic conditions, and demographics.

Psychological Vulnerabilities

Reactions to JI are not simply a function of whether individuals can replace what is lost. Rather, they are a function of the degree of potential loss from a psychological standpoint—in particular, the extent to which people have important identities, roles, and expectations tied to their job or its security. The final set of moderating conditions, *psychological vulnerabilities*, thus concerns the extent to which JI threatens important identities, roles, and expectations. Psychological vulnerabilities heighten stress associated with JI, given that JI poses an even greater threat to the self under these circumstances. Accordingly, research finds that people who

strongly identify with their jobs and who base a large part of their self-esteem on work-related achievement respond more negatively to JI (Blom, Richter, Hallsten, & Svedberg, 2015; Probst, 2000). Similar effects would be expected under conditions of high on-the-job embeddedness, which captures psychological and social investment in one's job (Crossley et al., 2007). Psychological vulnerabilities—in particular, expectations regarding job security—can affect the degree to which JI is experienced as an imbalance in the social exchange relationship. For example, research finds that those who have greater expectations regarding security perceive JI as more unfair, are less committed, and have greater intentions to quit (Bernhard-Oettel, De Cuyper, Schreurs, & De Witte, 2011). While those who are most psychologically vulnerable might be most motivated to preserve their jobs, stress-related effects may counteract these efforts (e.g., for findings regarding job-involved employees, see Probst, 2000).

Key drivers of psychological vulnerabilities. Several factors shape identities, roles, and expectations and would therefore be expected to affect reactions to JI. For example, past research reveals that JI is more distressing for men given the greater threat to their identity and traditional role as the breadwinner (J. Lam et al., 2014). However, women with egalitarian, as opposed to traditional, gender values experience similar responses to JI as do men (Gaunt & Benjamin, 2007). JI has been found to be more detrimental among employees with collectivist cultural values, as it threatens culturally valued group ties and security (reflecting greater psychological vulnerability; Probst & Lawler, 2006).

Elements of the formal and psychological contract are also relevant. Temporary employees engage in less relational contracting than permanent employees and are therefore less likely to view JI as a violation of their expectations (De Cuyper & De Witte, 2006). Thus, although employees in temporary positions experience greater JI, permanent employees experience more severe consequences in terms of satisfaction, commitment, and sickness absenteeism, reflecting their greater psychological vulnerabilities (De Cuyper & De Witte, 2006; Klandermans et al., 2010). The greater their investments into their positions (e.g., in terms of tenure, sacrifices made for the job), the more security individuals might expect and the more likely it is that they will see JI as unjust or a breach of expectations. In contrast, some have argued that an expansive view of one's career (i.e., a "boundary-less" career orientation) can enable flexibility and reduce negative reactions to JI by lowering expectations for security (Klehe et al., 2012; Otto et al., 2010). With this regard, personality is expected to be relevant, as traits shape expectations, goals, and values. In summary, to the extent that JI threatens key components of one's identity, sources of esteem, and expectations, people are psychologically vulnerable to greater detrimental reactions to JI.

Summary, Implications, and Further Directions

This framework provides an accounting of key variables that would be expected to affect reactions to JI. In doing so, it provides a rich perspective on who is most adversely affected by JI and why. These moderating factors further suggest several avenues through which organizations and policy makers can intervene to affect reactions to JI, and they pave the way for future work to uncover additional moderators. They also suggest particular challenges for organizations, as those most predisposed to provide effort and loyalty (i.e., high job involvement) are at greatest risk for negative well-being and attitudinal consequences of JI. Viewed in conjunction with the previous discussion of antecedents, this framework contributes to

research and practice by providing a means to identify particularly perilous conditions: those variables (notated in Table 3) that not only enhance JI but also predict more harmful reactions by virtue of their impact on one or more moderating conditions. For instance, by virtue of experiencing both greater JI and greater economic vulnerabilities, certain groups (e.g., lower educated, blue-collar) may be particularly disadvantaged (Kalleberg, 2011).

Beyond investigating the ideas set forth here, future work might expand the framework in several directions. First, future work should examine whether effects differ according to quantitative versus qualitative JI. For example, economic vulnerabilities might play a larger role in shaping reactions to quantitative JI since they are more focused on the job as a whole. Second, future research might extend the model to examine JI as a collective construct. Emerging research suggests that JI climate (shared perceptions of JI) influences outcomes above and beyond individual JI and can exacerbate the effects of individual JI (e.g., Jiang & Probst, 2016; Sora, De Cuyper, Caballer, Peiró, & De Witte, 2013). Third, and relatedly, research might expand the framework to examine group- and organization-level outcomes (e.g., Shoss & Probst, 2012).

Conclusions

This article provides an overview of the rapidly growing JI literature aimed at identifying crucial gaps and providing a general framework through which to guide future work. To that end, it calls for greater attention to the conceptualization and measurement of JI, specifically to eliminate restrictive construct assumptions and measurement confounds. Research is additionally needed to expand knowledge of antecedents of JI, in particular by recognizing that JI occurs not only when individuals perceive that future of the job itself is threatened but also when they perceive certain risks to themselves as jobholders. The proposed framework calls for greater attention to ways that employees try to counteract threats and proactively cope with potential loss, as well as the role that moderators play in shaping these and other reactions to JI. These questions demand longitudinal work to more precisely test hypotheses (e.g., regarding increases or decreases in attitudes or behaviors) and explore feedback loops. By laying out key gaps needing research attention, the discussion provided here will ideally help stimulate research that produces greater insights into this increasingly frequent and important experience.

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